

EIGRP PACKET FLOW:

EIGRP uses several packet types for its operations:

Hello Packets: Used to discover and maintain neighbor relationships.

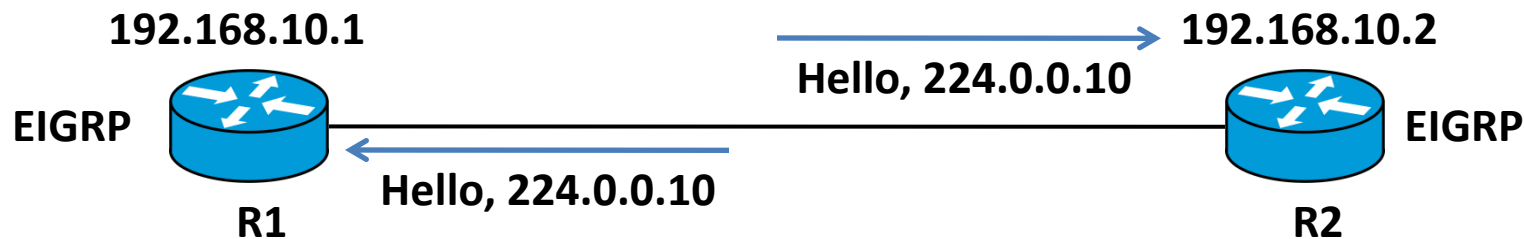
Update Packets: Used to propagate routing information.

Query Packets: Used to request routing information from neighbors.

Reply Packets: Used to respond to query packets.

Acknowledgment Packets: Used to acknowledge the receipt of update, query, and reply packets.

Hello Packet for Neighbor Discovery: : EIGRP



The Hello packet will be exchanged in between Routers when the EIGRP Configured

Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0]

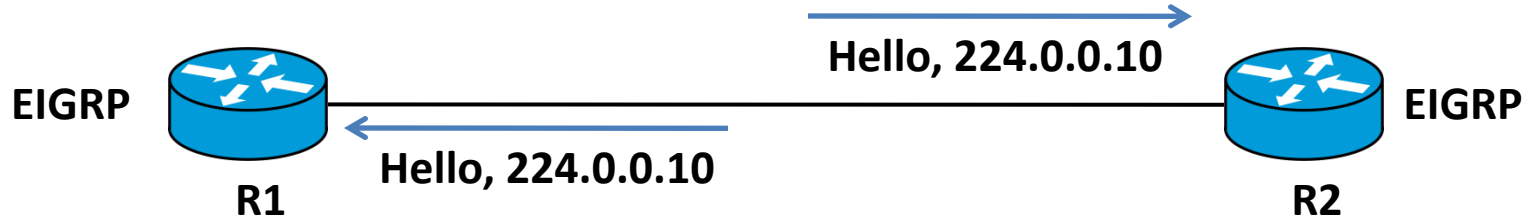
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No.	Time	Source	Destination	Protocol	Length	Info
23	68.633846	ca:02:3f:cc:00:08	ca:02:3f:cc:00:08	LOOP	60	Reply
24	71.441333	192.168.10.2	224.0.0.10	EIGRP	74	Hello
25	71.456320	192.168.10.1	224.0.0.10	EIGRP	84	Hello
26	71.471281	ca:01:48:c0:00:08	Broadcast	ARP	60	Who has 192.168.10.2?
27	71.502208	ca:02:3f:cc:00:08	ca:01:48:c0:00:08	ARP	60	192.168.10.2 is at ca:
28	71.502208	192.168.10.2	224.0.0.10			

<https://www.youtube.com/@techwithravish>

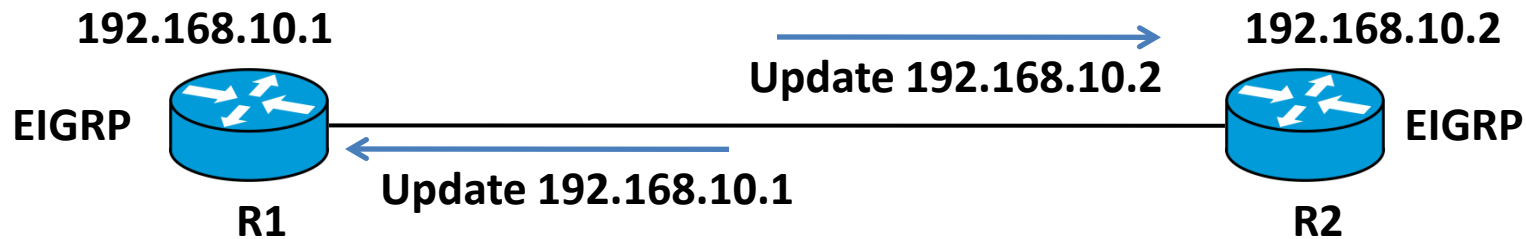
Neighbor Table will be Updated : EIGRP



```
⋮ R1 x R2 +
```

```
*Jul 28 13:13:04.159: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip ei
R1#show ip eigrp ne
R1#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H   Address                Interface      Hold Uptime    SRTT   RTT  Q  Seq
                               (sec)          (ms)                Cnt  Num
0   192.168.10.2            Fa0/0         11 00:05:25    1   3000  0   1
R1#
R1#
R1#
R1#
```

Update Packet will be forwarded : EIGRP



Update Message will exchange routing information in between routers

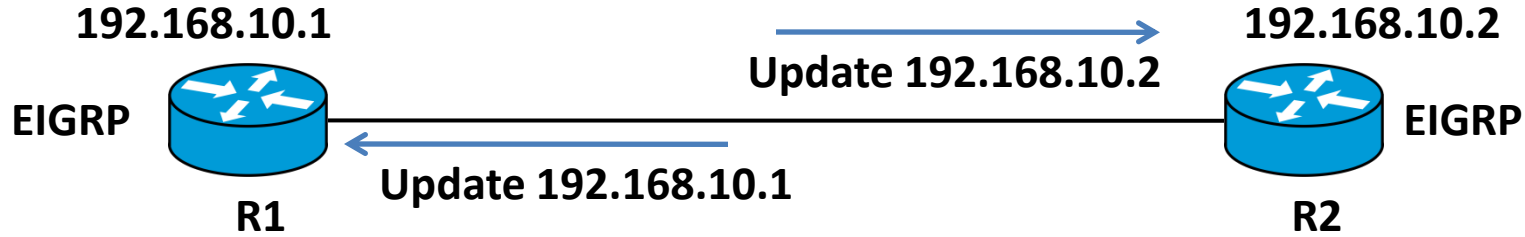
Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0]

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No.	Time	Source	Destination	Protocol	Length	Info
27	71.502208	ca:02:3f:cc:00:08	ca:01:48:c0:00:08	ARP	60	192.168.10.2 is at ca:
28	71.502208	192.168.10.2	224.0.0.10	EIGRP	84	Hello
29	71.534122	192.168.10.2	192.168.10.1	EIGRP	60	Update
30	73.478894	192.168.10.1	192.168.10.2	EIGRP	60	Update
31	73.509810	192.168.10.2	192.168.10.1	EIGRP	60	Hello (Ack)

Topology Table will be updated : EIGRP

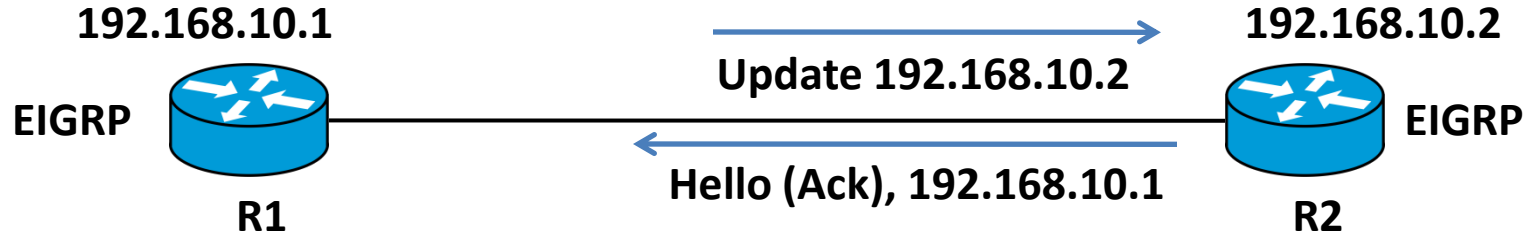


```
R1#
R1#
R1#show ip ei
R1#show ip eigrp da
R1#show ip eigrp to
R1#show ip eigrp topology
EIGRP-IPv4 Topology Table for AS(1)/ID(192.168.10.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 192.168.10.0/24, 1 successors, FD is 28160
   via Connected, FastEthernet0/0

R1#
R1#
```

Acknowledgement Message: EIGRP



In response of Update Message Acknowledgement will be forwarded

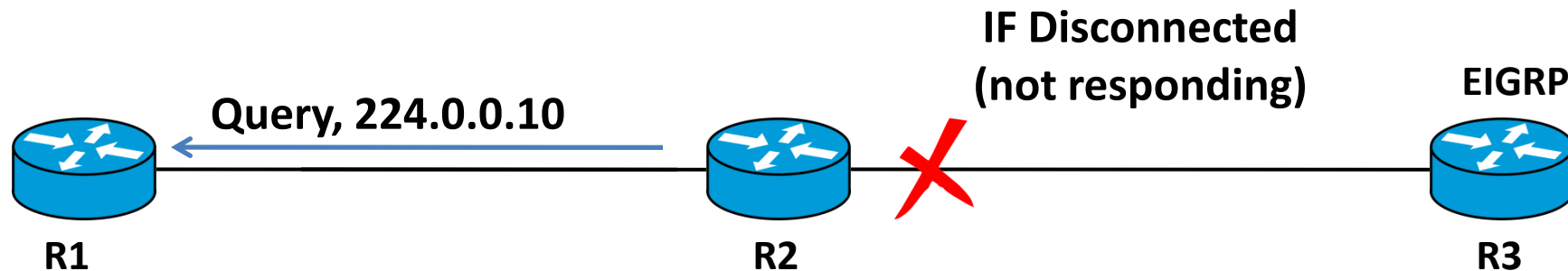
Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0]

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No.	Time	Source	Destination	Protocol	Length	Info
27	71.502208	ca:02:3f:cc:00:08	ca:01:48:c0:00:08	ARP	60	192.168.10.2 is at ca:
28	71.502208	192.168.10.2	224.0.0.10	EIGRP	84	Hello
29	71.534122	192.168.10.2	192.168.10.1	EIGRP	60	Update
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31	73.509810	192.168.10.2	192.168.10.1	EIGRP	60	Hello (Ack)

Query Message: EIGRP



Query message generated to find alternative path

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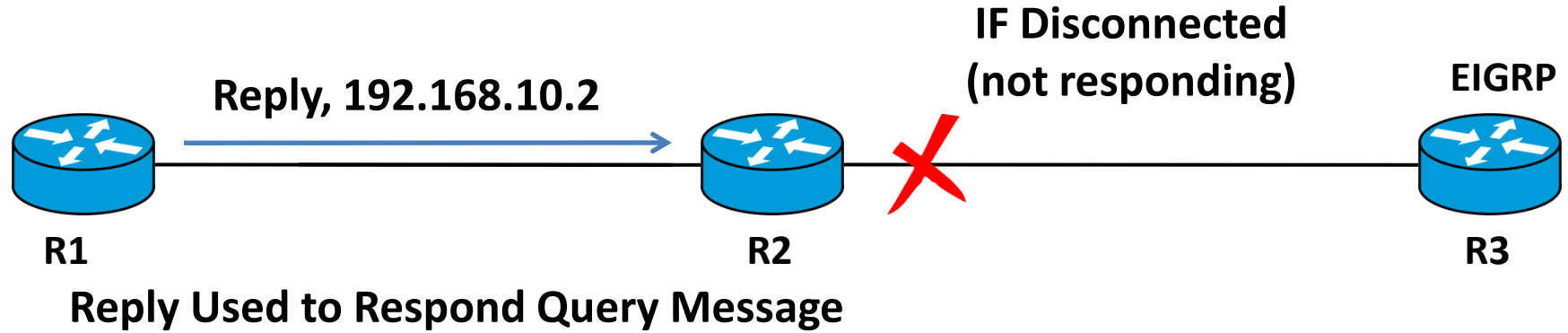
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No.	Time	Source	Destination	Protocol	Length	Info
1218	1837.047065	ca:02:3f:cc:00:08	CDP/VTP/DTP/PAgP/UDLD	CDP	366	Device ID: R2 Port ID
1219	1837.077982	192.168.10.2	224.0.0.10	EIGRP	98	Query
1220	1837.092943	192.168.10.1	192.168.10.2	EIGRP	60	Hello (Ack)
1221	1837.107904	192.168.10.1	192.168.10.2	EIGRP	98	Reply
1222	1837.139818	192.168.10.2	192.168.10.1	EIGRP	60	Hello (Ack)
1223	1837.632531	192.168.10.1	224.0.0.10	EIGRP	74	Hello

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Reply Message: EIGRP



Capturing from - [R1 FastEthernet0/0 to R2 FastEthernet0/0]

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1218	1837.047065	ca:02:3f:cc:00:08	CDP/VTP/DTP/PAgP/UDLD	CDP	366	Device ID: R2 Port ID
1219	1837.077982	192.168.10.2	224.0.0.10	EIGRP	98	Query
1220	1837.092943	192.168.10.1	192.168.10.2	EIGRP	60	Hello (Ack)
1221	1837.107904	192.168.10.1	192.168.10.2	EIGRP	98	Reply
1222	1837.139818	192.168.10.2	192.168.10.1	EIGRP	60	Hello (Ack)
1223	1837.632531	192.168.10.1	224.0.0.10	EIGRP	74	Hello

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